

CAMPUS CARBON (Version 2.0)

Universal Campus Carbon Footprint Assessment & Management Platform

CONSOLIDATED EXECUTIVE ASSESSMENT & SUSTAINABILITY PERFORMANCE REPORT

1. Assessment Metadata & Institutional Profile

Institution Name:	Christ University – Kengeri Campus	Reporting Year:	2025 (Annual Baseline)
Location:	Kumbalgodu, Mysuru Rd, Bengaluru, India	Standard:	GHG Protocol / ISO 14064-1
Total Campus Area:	78.50 Acres (317,678.20 m²)	Built-up Area:	55.0% = 43.18 Acres (174,750.0 m²)
Campus Population:	6,450 (6,000 Students, 450 Staff)	Buildings Inventory:	19 Operational Buildings
Grid Emission Factor:	India CEA v19 (0.716 kg CO ₂ e/kWh)	Operational Boundary:	100% Operational Control (All Scopes)

2. Five Core Carbon Accounting Pillars

GROSS FOOTPRINT 1,796.54 tCO ₂ e (Total Positive)	NEGATIVE REDUCTIONS -640.01 tCO ₂ e (Avoided & Sinks)	NET CARBON FOOTPRINT 1,156.54 tCO ₂ e (Gross - Reductions)	PER CAPITA INTENSITY 0.1793 tCO ₂ e / person (179.3 kg)	AREA INTENSITY 6.62 kg CO ₂ e / m² built-up
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3. GHG Inventory Scope & Activity Breakdown

Scope	Emission Source / Stream	Annual Quantity	Emissions (tCO ₂ e)	% Share
Scope 2	Grid Electricity (BESCOM Supply)	1,850,000 kWh	1,324.60	73.73%
Scope 1	Backup DG Generators (Diesel)	42,000 Liters	112.56	6.27%
Scope 1	Campus Transport Buses (Diesel)	55,000 Liters	147.40	8.20%
Scope 1	Dining Hall Kitchens (LPG)	26,000 kg	77.84	4.33%
Scope 1	Campus EV Buggy Shuttles (2 units)	4,800 kWh	0.00*	0.00%*
Scope 3	Solid Waste to Municipal Landfill	28,000 kg	14.37	0.80%
Scope 3	Deep Borewell Water Pumping	145,000 kL	89.90	5.00%
Scope 3	Central SBR Sewage Treatment Plant	42,000 kWh	29.48	1.64%
TOTAL	Gross Carbon Footprint (Scope 1 + 2 + 3)	—	1,796.54	100.00%

* EV charging electricity is included in Scope 2 grid utility bill; tracked with zero Scope 1 charge to prevent double counting.

4. Negative Carbon Footprint Ledger (Avoided Emissions & Sinks)

Negative Activity Stream	Asset Description	Methodology & Standard	Reduction (tCO ₂ e)
■ On-Site Solar PV	500 kWp Rooftop Solar (720,000 kWh/yr)	Grid Displacement (0.716 kg/kWh)	-515.52
■ Greenery & Trees	2,800 Mature Trees across 182,000 m²	IPCC GPG / FSI (22.0 kg CO ₂ /tree/yr)	-61.60
■ Waste Diversion	82 t Composted + 45 t Recycled (81.9% rate)	Avoided Landfill Methane Standard	-55.39
■ Retention Ponds	2 Lakes / Retention Wetlands (15,000 m²)	Ramsar Sediment Carbon Stabilization	-7.50
TOTAL REDUCTIONS	Combined Removals & Avoided Emissions	—	-640.01

5. Campus Building Spatial & Demographic Inventory

Code	Building Name	Type	Floors	Area (m²)	Resident	Day Schl	Faculty	Staff	Total Pop
ENG-01	Engineering Block I (Civil & Mech)	Academic	5	14,500	50	500	30	20	600
ENG-02	Engineering Block II (CSE & ECE)	Academic	5	16,200	60	620	40	30	750
ARCH-01	School of Architecture & Planning	Academic	4	9,800	30	330	25	15	400
MGMT-01	School of Business & Management	Academic	4	11,500	40	460	30	20	550
ARTS-01	Social Sciences & Humanities	Academic	4	8,500	30	350	25	15	420
RES-01	Center for Research & Innovation	Laboratory	3	7,200	20	200	20	10	250
LIB-01	Central Knowledge & Library Center	Library	3	7,400	20	250	15	15	300
AUD-01	Main Auditorium & Cultural Complex	Auditorium	2	6,800	10	170	10	10	200
HST-01	St. Kuriakose Boys Hostel	Hostel	6	12,500	240	0	5	5	250
HST-02	Devadan Hall Boys Hostel	Hostel	6	11,800	210	0	5	5	220
HST-03	Carmel Hall Girls Hostel	Hostel	6	13,200	250	0	5	5	260
HST-04	Mother Teresa Girls Hostel	Hostel	6	12,800	200	0	5	5	210
QTR-01	Staff & Faculty Residential Quarters	Residential	4	8,200	0	0	50	40	90
SPRT-01	Indoor Sports Complex & Gym	Sports	2	6,500	10	120	10	10	150
DIN-01	Central Dining Hall & Food Court	Canteen	2	5,500	20	140	10	10	180
LAB-01	Advanced Computing & AI Labs	Laboratory	3	6,200	20	180	10	10	220
LAB-02	Robotics & IoT Workshop	Workshop	2	5,400	10	130	10	10	160
ADM-01	Central Administrative Block	Admin	4	7,650	0	180	20	40	240
MED-01	Campus Health Center & Clinic	Medical	2	7,200	0	70	15	15	100
TOTAL	19 Buildings Reconciled	—	—	174,750	900	5,100	300	150	6,450

6. Prioritized Decarbonization Roadmap & Net-Zero 2030 Actions

Intervention & Project Title	Action Type	Priority	Abatement (tCO ₂ e)	Payback
Solar PV Expansion (+500 kWp) Double rooftop solar capacity to 1,000 kWp on academic blocks.	Renewable Energy	High	-358.0 tCO ₂ e	3.8 Yrs
Campus Fleet Bus Electrification Transition 15 diesel buses to electric fleet with depot chargers.	Clean Mobility	High	-147.4 tCO ₂ e	4.2 Yrs
Deep HVAC & Smart LED Retrofits BLDC inverter split ACs and daylight-sensor corridor LEDs.	Energy Efficiency	High	-198.7 tCO ₂ e	2.1 Yrs
Miyawaki Native Forest Plantation Plant 1,200 high-density native trees (Neem, Peepal, Teak).	Sequestration	Medium	-26.4 tCO ₂ e	1.5 Yrs
Zero-Waste Closed-Loop Biogas Digester Convert 82 t/yr canteen food waste to clean cooking gas.	Waste-to-Energy	Medium	-31.2 tCO ₂ e	2.9 Yrs
PROJECTED TOTAL ABATEMENT	Net-Zero 2030 Transition	—	-761.7 tCO ₂ e	< 3.2 Yrs Avg

Assessment Certification: Prepared in accordance with ISO 14064-1 & GHG Protocol Corporate Accounting Standard. Verified by Campus Sustainability Cell.